

Relations

ความสัมพันธ์

Chapter Summary

1. Relations (Revisit)
2. Properties of Relations
3. n-ary Relations and Their Applications
4. Representing Relations

1. Relations

An **ordered pair** is a set of inputs and outputs and represents a relationship between the two values.

1. Relations (Previously before Midterm)

ความสัมพันธ์

- ความสัมพันธ์ (Relation) เป็นโครงสร้างที่ใช้บ่งบอกสัมพันธภาพ (Relationship) ระหว่างวัตถุต่าง ๆ ความสัมพันธ์เป็นเซตของ n -สิ่งอันดับ ซึ่งเซตดังกล่าวเป็นเซตย่อยของผลคูณคาร์ทีเซียนของเซต n เซต
 - $R1 = \{(x1, 1.5), (x2, 0.35), (y, 0.987)\}$ เป็นความสัมพันธ์ซึ่งบ่งบอกสัมพันธภาพระหว่างชื่อตัวแปรกับค่าของตัวแปรนั้น
 - $R2 = \{(\text{กร}, 0001, 02-218-9998), (\text{กิ่งแก้ว}, 0002, 02-271-5647), (\text{ขนิษฐา}, 0003, 089-001-0008), (\text{คมฤทธิ}, 0004, 085-134-8970)\}$ เป็นความสัมพันธ์ซึ่งบ่งบอกสัมพันธภาพระหว่างชื่อบุคคล รหัสประจำตัว และหมายเลขโทรศัพท์

1. Relations (Previously before Midterm)

ความสัมพันธ์

- ในกรณีที่ความสัมพันธ์เกิดขึ้นระหว่างเซตสองเซตเช่นความสัมพันธ์ซึ่งเป็นเซตย่อยของ $A \times B$ เราเรียกว่า ความสัมพันธ์ทวิภาค (Binary relation) จาก A ไป B จากนิยามนี้จะเห็นได้ว่าความสัมพันธ์นั้นเป็นกรณีทั่วไปของฟังก์ชัน
- มีความเป็นไปได้ที่จะเกิดความสัมพันธ์บนเซตใดเซตหนึ่ง (Relation on a set) เช่นความสัมพันธ์จาก A ไป A เราเรียกว่าความสัมพันธ์บนเซต A ซึ่งเป็นเซตย่อยของ $A \times A$ นั่นเอง

1. Relations (Previously before Midterm)

กิจกรรมกลุ่ม

- หากกำหนดให้ $A = \{1,2,3\}$ และ $B = \{2,3,4\}$ จงยกตัวอย่างความสัมพันธ์จาก A ไปยัง B และความสัมพันธ์บนเซต A

Cartesian Product

Definition: The Cartesian Product of two sets A and B , denoted by $A \times B$ is the set of ordered pairs (a, b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) | a \in A \wedge b \in B\}$$

Example 1:

$$A = \{a, b\} \quad B = \{1, 2, 3\}$$

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

Cartesian Product

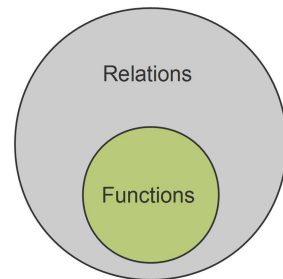
Definition: A subset R of the Cartesian product $A \times B$ is called a **relation** from the set A to the set B .

Relations

A **relation** is a set of inputs and outputs, and a **function** is a relation with one output for each input.

Some relationships make sense and others don't.

- Functions are relationships that make sense.
- **All functions are relations**, but not all relations are functions.

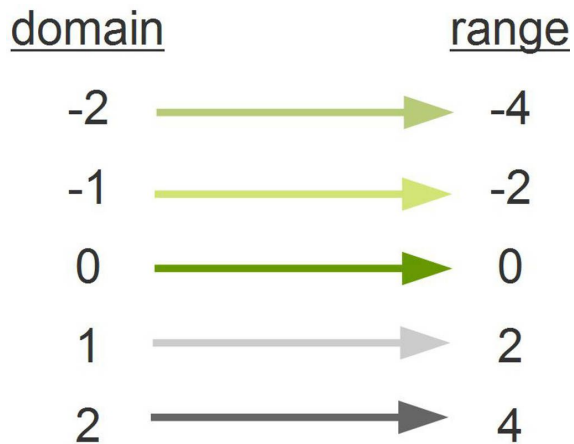


Functions as Relations

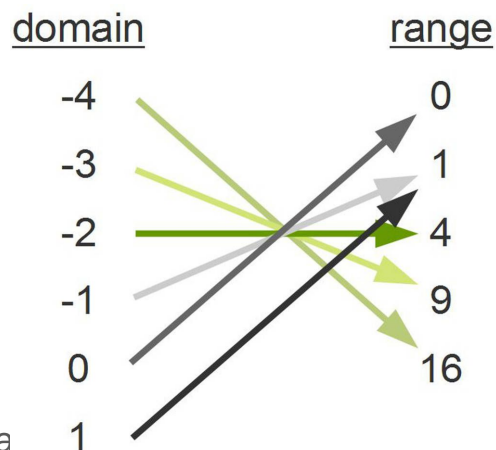
A function is a relation that for each input, there is only one output.

The **domain** is the **input** or the x-value, and the **range** is the **output**, or the y-value.

Each x-value is related to only one y-value.

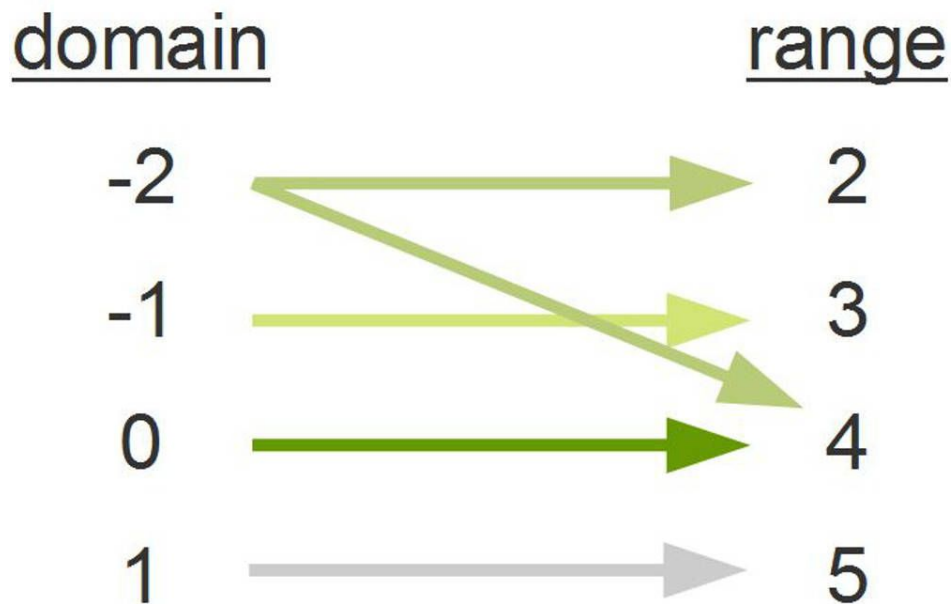


Dr. Sirasit Lochana



Functions as Relations

This mapping is **not a function**. The input for -2 has more than one output.



Binary Relations

Definition: A binary relation R from a set A to a set B is a subset $R \subseteq A \times B$.

In other words, a binary relation from A to B is a set R of ordered pairs, where the first element of each ordered pair comes from A and the second element comes from B .

Binary Relations on a Set

Definition: A binary relation R on a set A is a subset of $A \times A$ or a relation from A to A .

Example 1: Suppose that $A = \{a, b, c\}$.

- Then $R = \{(a, a), (a, b), (a, c)\}$ is a relation on A .

Binary Relations on a Set

Example 2: Let $A = \{1, 2, 3, 4\}$, which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

- Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b .

1 •	• 1	R	1	2	3	4
2 •	• 2	1				
3 •	• 3	2				
4 •	• 4	3				
		4				

Binary Relations on a Set

Example 3: Consider these relations on the set of integers:

- $R_1 = \{(a, b) \mid a \leq b\},$
- $R_2 = \{(a, b) \mid a > b\},$
- $R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$
- $R_4 = \{(a, b) \mid a = b\},$
- $R_5 = \{(a, b) \mid a = b + 1\},$
- $R_6 = \{(a, b) \mid a + b \leq 3\}.$

Which of these relations contain each of the pairs (1, 1), (2, 1), (1, -1)?

Binary Relations on a Set

Example 4: Consider these relations on the set of $A \times B$ where

$A = \{1, 3, 5\}$, $B = \{9, 11\}$:

- $R_1 = \{(1, 9), (5, 11)\}$
- $R_2 = \{(a, b) \mid a + 4 = b\}$,
- $R_3 = \{(a, b) \mid a \geq b\}$,

Which of these relations are valid?

2. Properties of Relations

1. Reflexive Relations

In some relations, an element is always related to itself.

Definition: R is reflexive if $(a, a) \in R$ for every element $a \in A$.

- Written symbolically, R on the set A is reflexive if and only if

$$\forall a[(a, a) \in R].$$

- A relation on A is reflexive if every element of A is related to itself.

Reflexive Relations

Example 1: Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R_6 = \{(3, 4)\}.$$

Which of these relations are reflexive?

Reflexive Relations

Example 2: Consider the following relations on $A = \{1, 2, 3\}$:

$$R_1 = \{(a, b) \in A \times A \mid a \leq b\},$$

$$R_2 = \{(a, b) \in A \times A \mid a < b\},$$

$$R_3 = \{(a, b) \in A \times A \mid a = b \wedge a < 3\},$$

Which of these relations are reflexive?

Reflexive Relations

Example 3: Consider these relations on the set of integers:

- $R_1 = \{(a, b) \mid a \leq b\},$
- $R_2 = \{(a, b) \mid a > b\},$
- $R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$
- $R_4 = \{(a, b) \mid a = b\},$
- $R_5 = \{(a, b) \mid a = b + 1\},$
- $R_6 = \{(a, b) \mid a + b \leq 3\}.$

Which of these relations are reflexive?

Properties of Relations

2. Symmetric Relations

In some relations, an element is related to a 2nd element if and only if the 2nd element is also related to the first element.

Definition: R is symmetric if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$.

- Written symbolically, R on the set A is symmetric if and only if

$$\forall a \forall b [(a, b) \in R \rightarrow (b, a) \in R].$$

- A relation on A is symmetric if and only if a is related to b always implies that b is related to a .

Symmetric Relations

Example 4: Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R_6 = \{(3, 4)\}.$$

Which of these relations are symmetric?

Symmetric Relations

Example 5: Consider these relations on the set of integers:

- $R_1 = \{(a, b) \mid a \leq b\},$
- $R_2 = \{(a, b) \mid a > b\},$
- $R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$
- $R_4 = \{(a, b) \mid a = b\},$
- $R_5 = \{(a, b) \mid a = b + 1\},$
- $R_6 = \{(a, b) \mid a + b \leq 3\}.$

Which of these relations are symmetric?

Properties of Relations

3. Antisymmetric Relations

In some relations, an element is related to a 2nd element, then this 2nd element is not related to the first.

Definition: R is antisymmetric if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ for all $a, b \in A$.

- Written symbolically, R on the set A is antisymmetric if and only if

$$\forall a \forall b [(a, b) \in R \wedge (b, a) \in R \rightarrow (a = b)].$$

- A relation on A is antisymmetric if and only if there are no pairs of distinct elements a and b with a related to b and b related to a .

Antisymmetric Relations

Example 6: Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R_6 = \{(3, 4)\}.$$

Which of these relations are antisymmetric?

Antisymmetric Relations

Example 7: Consider these relations on the set of integers:

- $R_1 = \{(a, b) \mid a \leq b\},$
- $R_2 = \{(a, b) \mid a > b\},$
- $R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$
- $R_4 = \{(a, b) \mid a = b\},$
- $R_5 = \{(a, b) \mid a = b + 1\},$
- $R_6 = \{(a, b) \mid a + b \leq 3\}.$

Which of these relations are antisymmetric?

Properties of Relations

4. Transitive Relations

In some relations, an element is related to a 2nd element, then this 2nd element is related to the 3rd element. Therefore, the first element is also related to the 3rd element.

Definition: R is transitive if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$ for all $a, b, c \in A$.

- Written symbolically, R on the set A is transitive if and only if

$$\forall a \forall b \forall c [(a, b) \in R \wedge (b, c) \in R \longrightarrow (a, c) \in R].$$

Transitive Relations

Example 8: Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R_6 = \{(3, 4)\}.$$

Which of these relations are transitive?

Transitive Relations

Example 9: Consider these relations on the set of integers:

- $R_1 = \{(a, b) \mid a \leq b\},$
- $R_2 = \{(a, b) \mid a > b\},$
- $R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$
- $R_4 = \{(a, b) \mid a = b\},$
- $R_5 = \{(a, b) \mid a = b + 1\},$
- $R_6 = \{(a, b) \mid a + b \leq 3\}.$

Which of these relations are transitive?

Equivalence Relations

Definition: A relation on a set A is called an *equivalence relation* if it is reflexive, symmetric, and transitive.

Equivalence Relation

Example 10: Consider these relations on the set of integers:

- $R_1 = \{(a, b) \mid a \leq b\},$
- $R_2 = \{(a, b) \mid a > b\},$
- $R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$
- $R_4 = \{(a, b) \mid a = b\},$
- $R_5 = \{(a, b) \mid a = b + 1\},$
- $R_6 = \{(a, b) \mid a + b \leq 3\}.$

Which of these relations are equivalence relation?

Equivalence Classes

Definition: Let R be an equivalence relation on a set A .

The set of all elements that are related to an element a of A is called the *equivalence class* of a .

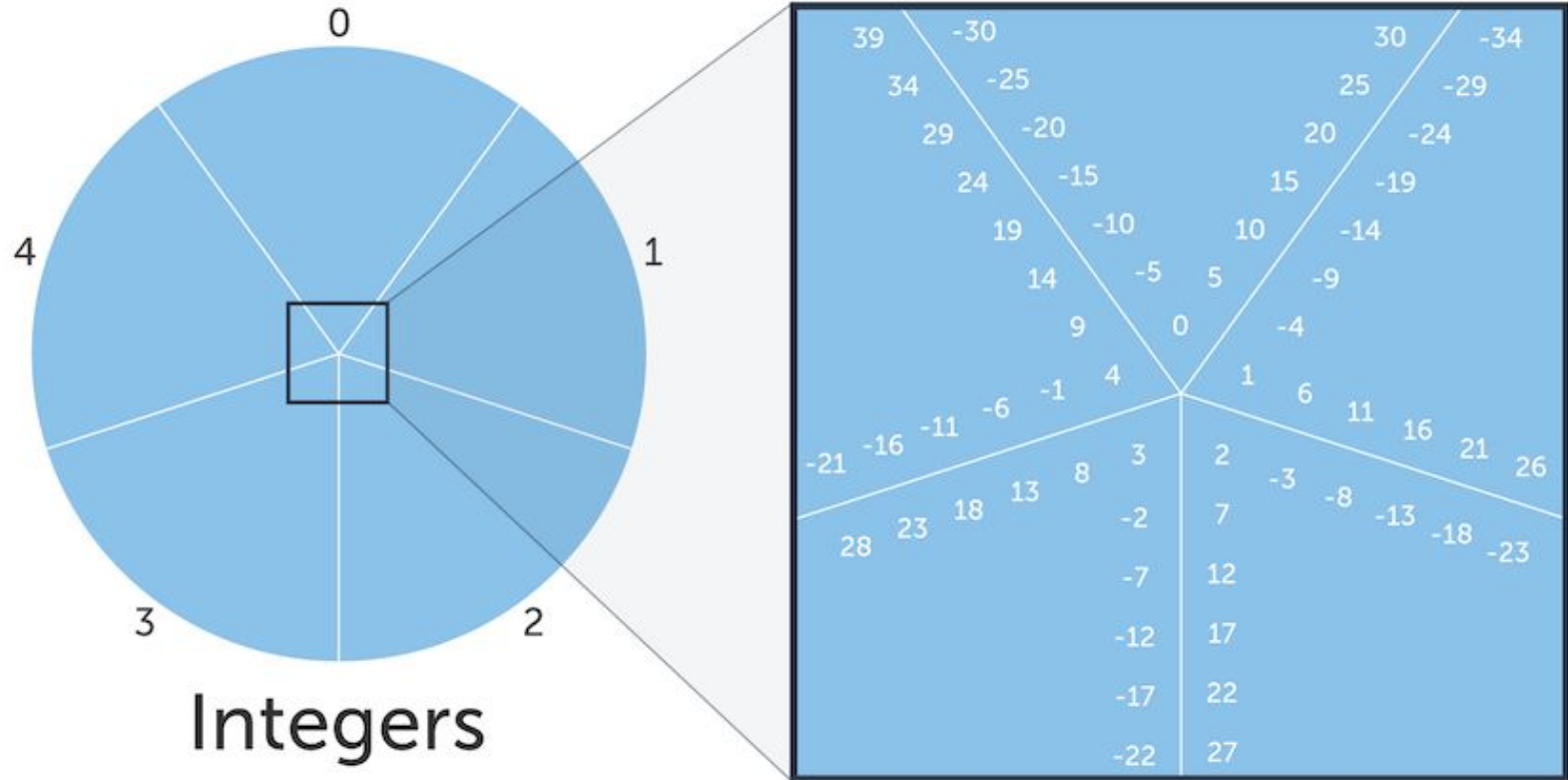
- The equivalence class of a with respect to R is denoted by $[a]_R$.
- The equivalence class of the element a is $[a]_R = \{s \mid (a, s) \in R\}$.
- If $b \in [a]_R$, then b is called a **representative** of this equivalence class.
 - Any element of a class can be used as a representative of this class.

Equivalence Classes

Example 11: Let R be the equivalence relation on the set of integers such that $(a, b) \in R$ if and only if $a = b$ or $a = -b$.

What are the equivalence classes for R ?

Congruence Modulo



Congruence Modulo is an Equivalence Relation

- Every pair of values in a slice are related to each other.
- No any value is in more than one slice.
- A pie contains all slices, which have **all** of the **integer** values.
- A pie has **equivalence relation**, which defines how to partition set of values (**mod**) into **congruence/equivalence** classes (slices).

Importance of Relation Properties

- Relations describe how objects are connected.
- Properties (reflexive, symmetric, antisymmetric, transitive) help us understand the **shape of connections** by recognize pattern in relationships.
- They are the foundation for reasoning, sorting algorithms, networking and classification in mathematics and computer science.

Relation Properties Example

1. Reflexive (every element relates to itself)
 - **Identity:** In a university's database, every every student has their own student ID.
 - Any ID (Primary key) is always “equal to itself”
 - Ensures no duplicate rows, each row is its own identity.
2. Symmetric (mutual relationship)
 - **Undirected network:** Facebook friendship
 - Instagram follows?

Relation Properties Example

3. Antisymmetric (**hierarchy** / one-way unless identical)

- If “Course A is a prerequisite of Course B” and “Course B is a prerequisite of Course A”, the only way both can be true is if $A = B$.
 - It prevents circular hierarchies
- **Data Structures:** Sorting Algorithms ($\leq$)

4. Transitive (pathfinding, chains of logic, dependency graphs)

- **Reasoning & Inference:** In a flight route, if there's a route from $A \rightarrow B$ and $B \rightarrow C$, then there's an indirect route $A \rightarrow C$.
 - Without transitivity, navigation apps wouldn't work.

Relation Properties Example

5. Equivalence Relations

- Group elements into **equivalence classes**
 - Modular arithmetic → cryptography, computer clocks.
 - File compression → group identical patterns.
 - Databases → grouping duplicate records.
 - Machine learning → clustering similar items.
- Basis for **classification**, clustering, partitioning sets into “non-overlapping buckets”

2. n -ary Relations

There are many relationships among elements of more than 2 sets.

- For instance, there is a relationship involving the name of a student, the student's major, and the student's grade point average (GPA).
- Similarly, there is a relationship involving the airline, flight number, departure port, destination, departure time, and arrival time of a flight.
- **n -ary relations** are used to represent databases.

n -ary Relations

Definition: Let $A_1, A_2, \dots, A_n$ be sets.

An n -ary relation over these sets is a subset of $A_1 \times A_2 \times \dots \times A_n$.

The sets $A_1, A_2, \dots, A_n$ are called the *domains* of the relation, and n is called its *degree*.

Ordered n-tuples

Example

2-tuples: (1, 2)

3-tuples: (7, 8, 9)

The *ordered n-tuple* $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element and a_2 as its second element and so on until a_n as its last element.

- 2-tuples are called *ordered pairs*.
- Two n -tuples are equal if and only if their corresponding elements are equal.
 - For example, the ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$.

n -ary Relations

Example 1: Let R be the relation on $N \times N \times N$ consisting of triples (a, b, c) , where a, b , and c are integers with $a < b < c$.

Then $(1, 2, 3) \in R$,

but $(2, 4, 3) \notin R$.

The degree of this relation is 3.

Its domains are all equal to the set of natural numbers (N).

n -ary Relations

Example 2: Let R be the relation consisting of 5-tuples (A, N, S, D, T) representing airplane flights, where

A is the airline,

N is the flight number,

S is the starting point,

D is the destination, and

T is the departure time.

(Thai Airways, 924, BKK, MUC, 15:00)

For instance, Thai Airways has flight 924 from BKK to MUC at 15:00.

The degree of this relation is 5, and its domains are the set of all airlines, the set of flight numbers, the set of cities, the set of cities (again), and the set of times.

Databases and Relations

A database consists of **records**, which are n -tuples, made up of **fields**.

The fields are the entries of the n -tuples.

- The **relational data model** represents a database of records as an n -ary relation.
- Relations used to represent databases are called **tables**.
- A domain of an n -ary relation is called a **primary key** when the value of the n -tuple from this domain determines the n -tuple.
 - No 2 n -tuples in the relation have the same value from this domain.

Databases and Relations

Name	ID	Major
Ash	231455	IT
Blue	888323	IT
Green	102147	DSBA

$R = \{(Ash, 231455, IT), (Blue, 888323, IT), (Green, 102147, DSBA)\}$

3. Representing Relations

Apart from table, there are 2 alternative methods for representing (binary) relations.

- Zero-one Matrices
- Directed graphs

Representing Relations Using Matrices

A relation between finite sets can be represented using a zero-one matrix.

Suppose R is a relation from $A = \{a_1, a_2, \dots, a_m\}$ to $B = \{b_1, b_2, \dots, b_n\}$.

The relation R is represented by the matrix $M_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

The matrix representing R has a 1 as its (i, j) entry when a_i is related to b_j and a 0 if a_i is not related to b_j .

Representing Relations Using Matrices

Example 1: Suppose that $A = \{1, 2, 3\}$ and $B = \{1, 2\}$.

Let R be the relation from A to B containing (a, b) if $a \in A$, $b \in B$, and $a > b$. What is the matrix representing R ?

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} & \\ & \\ & \end{bmatrix} \end{matrix}.$$

Matrices of Relations on Sets

- If R is a reflexive relation, all the elements on the main diagonal of M_R are equal to 1.

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{bmatrix}$$

- R is a symmetric relation, if and only if $m_{ij} = 1$ whenever $m_{ji} = 1$.
- R is an antisymmetric relation, if and only if $m_{ij} = 0$ or $m_{ji} = 0$ when $i \neq j$.

$$\begin{bmatrix} & & 1 \\ & \diagdown & \\ 1 & & \\ & \diagup & 0 \\ & 0 & \end{bmatrix}$$

$$\begin{bmatrix} & & 1 \\ & \diagdown & 0 \\ 0 & & \\ & \diagup & 0 \\ & 0 & 1 \end{bmatrix}$$

Matrices of Relations on Sets

Example 2: Suppose that the relation R on a set is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Is R reflexive, symmetric, and/or antisymmetric?

Representing Relations Using Digraphs

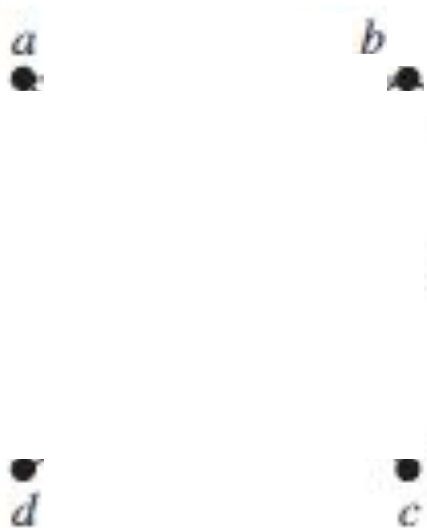
Definition: A *directed graph*, or *digraph*, consists of a set V of *vertices* (or *nodes*) together with a set E of ordered pairs of elements of V called *edges* (or *arcs*).

The vertex a is called the *initial vertex* of the edge (a, b) , and the vertex b is called the *terminal vertex* of this edge.

An edge of the form (a, a) is called a *loop*.

Representing Relations Using Digraphs

Example 3: A drawing of the directed graph with vertices a , b , c , and d , and edges (a, b) , (a, d) , (b, b) , (b, d) , (c, a) , (c, b) , and (d, b) .



Relations Properties Using Digraphs

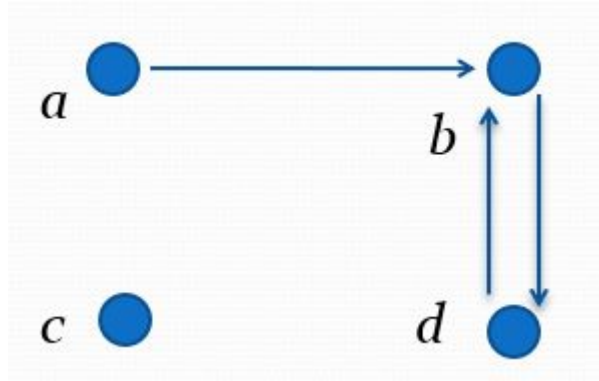
Reflexivity: A loop must be present at all vertices in the graph.

Symmetry: If (x, y) is an edge, then so is (y, x) .

Antisymmetry: If (x, y) with $x \neq y$ is an edge, then (y, x) is not an edge.

Transitivity: If (x, y) and (y, z) are edges, then so is (x, z) .

Relations Properties Using Digraphs Example



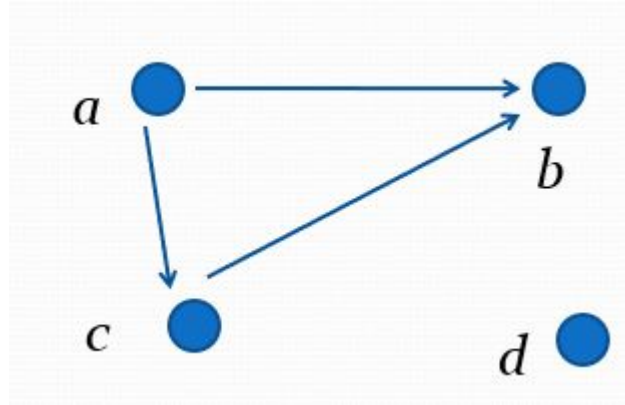
Reflexivity?

Symmetry?

Antisymmetry?

Transitivity?

Relations Properties Using Digraphs Example



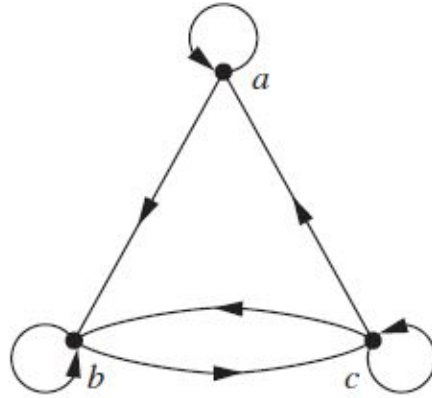
Reflexivity?

Symmetry?

Antisymmetry?

Transitivity?

Relations Properties Using Digraphs Example



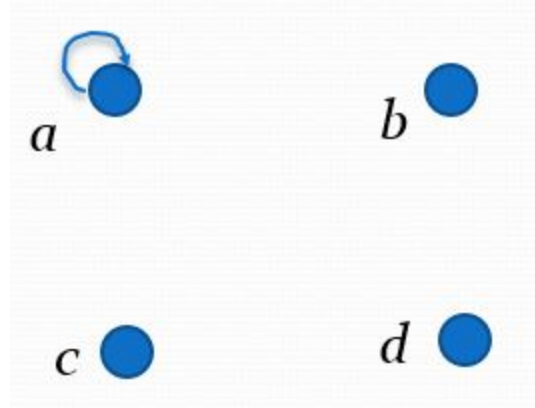
Reflexivity?

Symmetry?

Antisymmetry?

Transitivity?

Relations Properties Using Digraphs Example



Reflexivity?

Symmetry?

Antisymmetry?

Transitivity?